

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2080 Baishakh

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX 716)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. How is microwave frequency band classified by the IEEE? Enumerate the basic advantages and disadvantages of microwaves compared to lower frequencies. [2+4]
2. Design a double stub shunt tuner to match a load impedance of $Z_L = 60 - j80\Omega$ to a 50Ω line. The stubs are to be open circuited stubs and are spaced $3\lambda/8$ apart. Assume the first is 0.4λ from the load. Formulate the S-matrix for your design. [8+2]
3. Why are S-parameters used in microwave network analysis? Derive the S-parameters for a two port network. [2+6]
4. For an air-filled rectangular waveguide with a width of 3 cm and a desired frequency of operation of 6 GHz (for dominant mode), determine cut-off frequency, cut-off wavelength, group velocity, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. Explain the four basic parameters used to describe the performance of a directional coupler. [6+4]
5. Sketch a cross-sectional view of a magnetron having 45 degrees of phase shifts among the adjacent cavities, and explain its functioning as a power amplifier. Explain the bunching effect. [6+2]
6. Investigate the stability of a transistor having following S-parameters at 6GHz. [5+5]

$$[S] = \begin{bmatrix} 0.894\angle -60.6^\circ & 0.020\angle 62.4^\circ \\ 3.122\angle 123.6^\circ & 0.781\angle -27.6^\circ \end{bmatrix}$$
7. Explain why insertion loss technique is used to design microwave filters. With proper labeling sketch microwave double-section shunt – arm types microwave LPF using micro strips. [3+5]
8. Discuss the possible RF radiation fields to the public exposer. Explain, what are the international standards and recommended (SARPs) practices to safe from such radiation. [3+3]
9. Explain power measurement using calorimeter wattmeter. What are the limitations of using single bridge bolometer? [6+2]
10. Write short notes on: [2×3]
 - a) MASER
 - b) E-plane Tee

Supplied Formulas:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}S_{21}|}$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$G_{T \text{ Max}} = \frac{1}{1 - |\Gamma_s|^2} \cdot |S_{21}|^2 \cdot \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

$$G_{TU \text{ Max}} = \frac{1}{1 - |S_{11}|^2} \cdot |S_{21}|^2 \cdot \frac{1}{1 - |S_{22}|^2}$$

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1. Differentiate between lumped and distributed circuit analysis. List out the areas of application of microwave system. Provide detail protocols of two microwave communication systems. [3+3+4]
2. A 75Ω coaxial line is terminated with a complex load of $109 + j 120 \Omega$. Design a double-stub matching system using short-circuited coaxial lines. Prepare its S-matrix. [8+2]
3. Briefly explain the need of S-parameter and its properties in microwave analysis. Explain three port network parameters using an example of E-plane Tee. [4+4]
4. What is dominant mode in a waveguide? What are the dominant modes for rectangular and circular waveguides? Does a rectangular waveguide of width 2.254 cm support the propagation of a signal having frequency 6 GHz? [1+1+2]
5. Draw a neat diagram of a magic tee and derive its S-parameter. [4]
6. What is bunching effect? Briefly describe the construction and operation of Traveling Wave Tube (TWT). [2+6]
7. A GaAs MESFET has the following S-parameters measured with a 50Ω resistance at 10 GHz: [10]

$$S_{11} = 0.55 \angle -160^\circ \quad S_{12} = 0.04 \angle 10^\circ$$

$$S_{21} = 4.82 \angle 180^\circ \quad S_{22} = 0.45 \angle -30^\circ$$

Determine the stability and compare maximum power gains for unilateral and bilateral modes using supplied formulas.
8. How is a low pass filter is prototyped based on Butterworth or Chebyshev approximations and converted into other types of filters? Implement a second order high pass filter π -section using microstrips. [10]
9. Briefly describe power measurement and the working principle of double channel Bolometer Bridge method. [8]
10. Write short notes on: (Any Two) [2×4]
 - a) HERP
 - b) TE_{mn} Field Equations of rectangular waveguide
 - c) Spectrum Analyzer

Supplied Formulas:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}S_{21}|}$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$G_{T \max} = \frac{1}{1 - |\Gamma_s|^2} \cdot |S_{21}|^2 \cdot \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

$$G_{TU \max} = \frac{1}{1 - |S_{11}|^2} \cdot |S_{21}|^2 \cdot \frac{1}{1 - |S_{22}|^2}$$

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1. List out the uses of microwaves in communication systems, and provide detail communication protocols of any three of them. [2+6]
2. Classify microwave propagation fields of a GSM BTS and its radiation parameters. Explain what non-ionizing SAR is and how it becomes hazardous to human body. [4+4]
3. Prepare S-Matrices of a perfectly working E-Plane Tee, H-Plane Tee and a Duplexer, and explore their uses. [8]
4. Define the field equations and the basic characteristic equations of a rectangular waveguide working on TM_{11} mode. [8]
5. Design a double-stub impedance matching network for a load of $\Gamma = 0.64\angle 58^\circ$ connected to a 100 Ohm transmission line. Prepare the S-Matrix of its matched network. [8+2]
6. Calculate the maximum gains of both bilateral and unilateral models of a transistor having the S-parameters of $S_{11} = 0.55\angle -150^\circ$, $S_{12} = 0.04\angle 20^\circ$, $S_{21} = 2.82\angle 180^\circ$ and $S_{22} = 0.45\angle -30^\circ$. Sketch a flow chart that describes the microwave amplifier designing procedure. [6+3+3]
7. Explain the steps of microwave filter designing. Provide a sketch of double pad T-Type passive filter using a micro-strip. [7+3]
8. Explain the working principles of a cross-field magnetron having 60° of phase differences among the adjacent cavities. [8]
9. Write short notes on: (Any Two) [2×4]
 - a) Stability analysis of a microwave amplifier
 - b) Double-channel Bolometer
 - c) Sketch of an Immittance Chart

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|},$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21});$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{T_{\max}} = \left(\frac{1}{1 - |\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

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1. Compare the behavior of circuits for Low frequency/ Conventional and RF/ Microwave bands. Classify microwave frequency band and its application in major areas. [3+3]

2. A 50Ω lossless transmission line is required to be matched with the load admittance $0.00813 + j0.0065 \text{ } \Omega$, by a double-stub shunt tuner with separation of $3\lambda/8$ and the distance of the first stub from the load is 0.01λ . Calculate the length of each stub by using the smith chart. Write the s-parameter for the matched network. [8+2]

3. Why S-parameters are used in high frequencies? The S-matrix of certain microwave network is given as

$$S = \begin{bmatrix} 0.4 + j0.5 & j0.6 \\ j0.6 & 0.4 - j0.5 \end{bmatrix} \quad [3+1+1+1+2]$$

- a) Is the network reciprocal?
- b) Is the network lossless?
- c) What is the return loss at the input?
- d) If the input power to the network is 5 watts. What is the reflected power?

4. Provide the fundamental field and characteristic equations of a circular waveguide for TE mode. [10]

5. How is the output of conventional tubes reduced at microwaves due to inter-electrode capacitance, lead inductance and transit time effect? Explain about the construction and working principle of TWT. [2+6]

6. For transistor having following S-parameter $S_{11} = 0.894 \angle -60.6^\circ$, $S_{21} = 3.122 \angle 123.6^\circ$, $S_{12} = 0.020 \angle 62.4^\circ$, $S_{22} = 0.781 \angle -27.6^\circ$. Determine the stability and compare maximum power gains for bilateral and unilateral modes. [5+5]

7. How is a low pass filter prototype based on Butterworth approximation designed using insertion loss method? Implement a low pass filter π section using microstrips. [6+2]

8. Explain the RF/MW radiation hazards and its safety practices. [3+3]

9. List out the major RF/MW measurement parameters. How the VSWR of any microwave transmitter (In case of $VSWR > 10$) can be measured? Explain. [2+6]

10. Write short notes on: [2×3]

- a) Microwave Magic Tee
- b) Gunn-diode

Supplied Formulas:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}S_{21}|}$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$G_{T \text{ Max}} = \frac{1}{1 - |\Gamma_s|^2} \cdot |S_{21}|^2 \cdot \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

$$G_{TU \text{ Max}} = \frac{1}{1 - |S_{11}|^2} \cdot |S_{21}|^2 \cdot \frac{1}{1 - |S_{22}|^2}$$

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Subject: - RF and Microwave Engineering (EX752)

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- ✓ Assume suitable data if necessary.

1. How the circuit at seismic band is different from its RF/Microwave counterparts? Explain. [8]
2. A 75-ohm, coaxial line is terminated with a normalized complex load of $0.4 + j0.85$ ohms. Design a double-stub matching system using short-circuited coaxial line of 75-ohm characteristic impedance. Sketch the network using micro strip. [10+2]
3. a) Analyze a three-port directional coupler using S-parameters. [4]
b) Which of the passive microwave device is explained by this S-matrix. Judge the condition and explain its characteristics.

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix} \quad [6]$$

4. Derive the expression for the field strength for TM waves for a air-filled circular waveguide. Check the dominant mode in TE and TM modes. [8+2]
5. With neat circuit diagrams and relevant equations, explain the velocity modulation process and bunching in a multicavity reflex klystron. [10]
6. a) Refer the sketched smith chart (Fig.Q6) and analyze/synthesize the stabilities. Assume necessary parameters as desired. Mention all the steps. [5]
b) Describe the insertion loss method used for the filter designing. [5]
7. a) Discuss in detail the power measurement using circulating calorimeter. [5]
b) How microwave radiation becomes hazardous to human body? [5]
8. What do you understand by immittance chart? Sketch it. List out all duality parameters vital to designing microwave networks. [10]

$$|\Delta| = |S_{11} S_{22} - S_{12} S_{21}|$$

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12} S_{21}|}$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12} S_{21}|}$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12} S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}.$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$G_{TU_{\max}} = \frac{1}{1 - |S_{11}|^2} |S_{21}|^2 \frac{1}{1 - |S_{22}|^2}.$$

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1. Design a single short and open-circuited shunt matching network for a transmission line using Smith Chart by considering an output reflection coefficient $\Gamma_L = 0.5 \angle 51^\circ$ Ohm and surge impedance $Z_0 = 50$ Ohm. [8]

2. Identify and explain the properties of a microwave passive device having following S-Matrix. [8]

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

3. Sketch a flowchart for designing a microwave amplifier using a GaAsFET. Consider the following S-parameters and find maximum gain for both bilateral and unilateral model. Also using the calculated value of Γ_{in} and Γ_{out} trace Z_{in} and Z_{out} in the smith chart. [4+4+4+4]

$$[S] = \begin{bmatrix} 0.656 \angle 146.7^\circ & 0.122 \angle 46.1^\circ \\ 2.30 \angle 44.7^\circ & 0.172 \angle -117.1^\circ \end{bmatrix}$$

4. Synthesize stability parameters of input matching network for the attached sketched smith chart. [8]

5. Choose a proper microwave measurement tool to test an antenna as a DUT; and explain its working principles. [8]

6. Explain in detail the designing steps of microwave filters. Illustrate an example of passive HPF using microstrips. [6+4]

7. Express field equations of a rectangular waveguide for TM mode. [10]

8. Write short notes on: (any two) [6×2]

- i) Effect of SAR as microwave radiation hazards
- ii) Features of microwave frequency band
- iii) Backward Wave Oscillator
- iv) Microwave Cavity Resonators

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|},$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$R_I = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{Tmax} = \left(\frac{1}{|1 - S_{11}\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

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1. What are the advantages of using double stub matching over single stub matching? Explain the necessary steps for impedance matching of a load to a transmission line using double-stub matching network with an appropriate example. Use provided Smith Chart. [2+8]
2. Why are S-parameters used in microwave network analysis? Analyze three port network using S-parameters. [2+6]
3. Suppose there are two identical radar transmitters in equipment stock. A particular application required twice more input power to an antenna than either transmitter can deliver. Give your appropriate solution for the given problem with explanation using S-matrix. [6]
4. Describe the working principle and applications of cavity magnetron having phase shift of 45° between the cavities. [8]
5. Given S-parameters for microwave transistor amplifier: [12]

$$S_{11} = 0.78 \angle -113^\circ \quad S_{12} = 0.028 \angle 247^\circ$$

$$S_{21} = 2.60 \angle 76^\circ \quad S_{22} = 0.81 \angle -54^\circ$$

Determine the stability and compare maximum power gain for unilateral and bilateral modes using supplied formulas.
6. How is low pass filter implemented using microstrip? How are the low pass filter prototyped? [5+3]
7. What are the different types of electromagnetic radiation hazard? Illustrate radiation fields of a microwave oven. [4+4]
8. Choose a proper power measurement tool to measure power of a Airport Surveillance Radar. [8]
9. Write short notes on: [4×3]
 - i) Behavior of microwave bands
 - ii) Field equation of rectangular wave guide for TE mode
 - iii) Cavity Resonators

Supplied Formulas:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}S_{21}|}$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$G_{T \text{ Max}} = \frac{1}{1 - |\Gamma_s|^2} \cdot |S_{21}|^2 \cdot \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

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1. Classify signal frequency in different bands of waves and rays. What are the advantages and disadvantages of using microwave signal? [3+5]
2. By assuming a complex inductive load of an antenna which is mismatched with the line impedance of 78.0 ohm, design a double-stub short-circuited matching network. Show both electrical and physical connections. [8+2]
3. Why S-parameter is important in microwave network analysis? Write down the properties of a 3-port network. [4+4]
4. Suppose there are two identical radar transmitters and few passive devices in equipment stock. A particular application requires twice more input power to an antenna than either transmitter can deliver. As a RF engineer, give your appropriate solution for the above problem with necessary figures, mathematics and sufficient explanation. [8]
5. What do you mean by slow backward wave structure? Explain the construction and working principle of a LNA. [2+6]
6. Show a flow diagram that explain designing of an amplifier using a FET transistor. With self-defined parameters and the help of a smith chart define conditional stability of a microwave amplifier. [10]
7. Justify and describe how a microwave filter is designed using insertion loss method. [2+6]
8. Define major microwave measurement parameters and explain the working principle of a low microwave power measurement device. [8]
9. Write short notes on: (any two) [2x6]
 - a) RF/MW radiation hazards and safety practices
 - b) Directional Couplers
 - c) TE mode circular wave guide



640-15143

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX 752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. Differentiate between lumped and distributed circuit analysis. What are the uses of microwave bands? [4+2]
2. Assume an inductive load impedance is connected to a mismatched 50Ω transmission line. Find the size and placement of the matching stub that will remove all the standing waves and match load to the line. Use double stub shunt tuning short and open circuited stub. Draw its electrical diagram and physical connection. [8]
3. Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example. [4+4]
4. Choose a suitable passive microwave device to split power into half and explain its properties. [8]
5. Explain what is bunching effect. Explain the working principle of BWO with neat diagrams. [2+8]
6. Check the stability and find the maximum gain a transistor amplifier having $S_{11} = 0.64\angle -169^\circ$, $S_{12} = 0.03\angle 50^\circ$, $S_{21} = 10.11\angle 91^\circ$, $S_{22} = 0.22\angle -82^\circ$. Consider both bilateral and unilateral model. Modify the S-parameters if necessary. [12]
7. Describe insertion loss method of microwave filter design. Illustrate an example of a passive LPF using μ -strip. [8+2]
8. Describe the working principle of a network analyzer. [8]
9. Write short notes on: (Any two) [2×5]
 - a) Microwave Circulators
 - b) TM mode for rectangular waveguides
 - c) Microwave radiation hazards

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}||S_{21}|}$$

$$\Delta = (S_{11}S_{22} - S_{12}S_{21})$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$\text{Where, } B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{Tmax} = \left(\frac{1}{1 - |\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

$$\text{For unilateral mode } S_{12} = 0, \Gamma_S = S_{11}^* \text{ and } \Gamma_L = S_{22}^*$$

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. What are the advantages and disadvantages of microwaves over acoustic waves? [5]
2. What is admittance chart? A load impedance of $Z_L = 80 + j100$ is connected to a microstrip transmission line. Find the size and placement of the matching stub. Use single stub shunt tuning short and open stubs. [2+8]
3. Define the use of S-parameters for three-port analysis. Define the term return loss and insertion loss. [5+2]
4. What are waveguide junctions? Describe the operational principles of magic tee based on s-parameters. [3+3]
5. What is density modulation? Describe the working principle of a multi-cavity klystron oscillator. [2+7]
6. Justify that a transistor having following S-parameters $S_{11} = 0.894 \angle -60.6^\circ$, $S_{12} = 0.020 \angle 62.4^\circ$, $S_{21} = 3.122 \angle 123.6^\circ$ and $S_{22} = 0.781 \angle -27.6^\circ$ is conditionally stable while designing an amplifier. Considering unilateral model calculate maximum gain. [5+5]
7. How can you implement low pass filter using micro-strip? How they are prototyped? [3+5]
8. Describe how standing waves and microwave powers are measured with VSWR meter and low power measurement. [2+8]
9. Write short notes on: (any three) [3×5]
 - a) Dominant mode in waveguide
 - b) Circulators
 - c) LNA cavity device inserting loss method for filter designing
 - d) Insertion loss method for filter designing

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. What are the merits and demerits of microwaves compared to seismic wave? [5]
2. Compare RF radiation fields and explain how they are hazardous to living body. [5]
3. Sketch an immittance chart and compare the scales. [5]
4. Design a signal shunt and open matching networks using Smith Chart for a transmission line having surge impedance of 75 Ohm and load impedance of $78.27 + j60.93$ Ohm. Sketch the physical diagram considering microstrips. [4+4+2]
5. You are supposed to measure about 7.5 mW of microwave power. Choose a proper power measuring device and explain its working principle. [10]
6. What is cross-field effect? Design and describe the working principle of a cross field cavity device for power amplifier. Consider a 90° phase shift between adjacent cavities. [3+7]
7. Describe magic Tee based on S-parameters. Differentiate between dominant and degenerate modes. Consider a rectangular waveguide having dimension of $a = 3b$ and find the dominant mode among TM_{01} , TM_{10} , TM_{11} , TM_{21} , TM_{12} , TM_{02} and TM_{20} . [6+2+2]
8. A GaAs FET transistor has the S-Parameters at 5 GHz with 50 Ohm line measured as $S_{11} = 0.45\angle 163^\circ$, $S_{12} = 0.04\angle 40^\circ$, $S_{21} = 2.55\angle -106^\circ$ and $S_{22} = 0.46\angle -65^\circ$. Check the stability and find the maximum power gain using this transistor. [10]
9. Write short note on: (any three) [3×5]
 - a) Microwave filter parameters and LPF prototyping
 - b) TM mode rectangular wave guide
 - c) Bunching effect in reflex klystron
 - d) Microwave mixer

Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Necessary formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. Describe field equations and other related parameters of a rectangular waveguide in TM mode. Compare TE₁₀ and TE₂₀ in terms of cut-off frequency and dominant mode. [8+2]
2. Design a double stub matching network using three-eighths wavelength $\left(\frac{3\lambda}{8}\right)$ separation that match an antenna having load of 300+j300 Ohm connected to a 300 Ohm transmission line. Justify your design. [10]
3. What is bunching effect? Describe the working principle of a klystron oscillator. [2+8]
4. Using the given S-parameters $S_{11}=0.55\angle 150^\circ$, $S_{12}=0.04\angle 20^\circ$, $S_{21}=2.82\angle 180^\circ$, $S_{22}=0.45\angle -30^\circ$ and required assumptions, calculate maximum gains of this transistor amplifier for bilateral and unilateral modes. [10]
5. Draw a flow diagram to describe the design procedure of a microwave amplifier. Define the stability of an amplifier having $C_S=1.15\angle 10^\circ$, $R_S=0.85$, $C_L=1.10\angle 80^\circ$, $R_L=1.10$. [5+5]
6. How microwave measurements are different to low frequency measurements? Describe how static calorimeter works to measure power. [3+7]
7. Design a two-port network model and derive the required parameters. [10]
8. Write short notes (Any TWO) [2x5]
 - a. Design procedures of microwave filters
 - b. Microwave radiation hazards and safety practices
 - c. Backward Wave Oscillator
 - d. Merits of S-parameters in microwaves

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|}$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

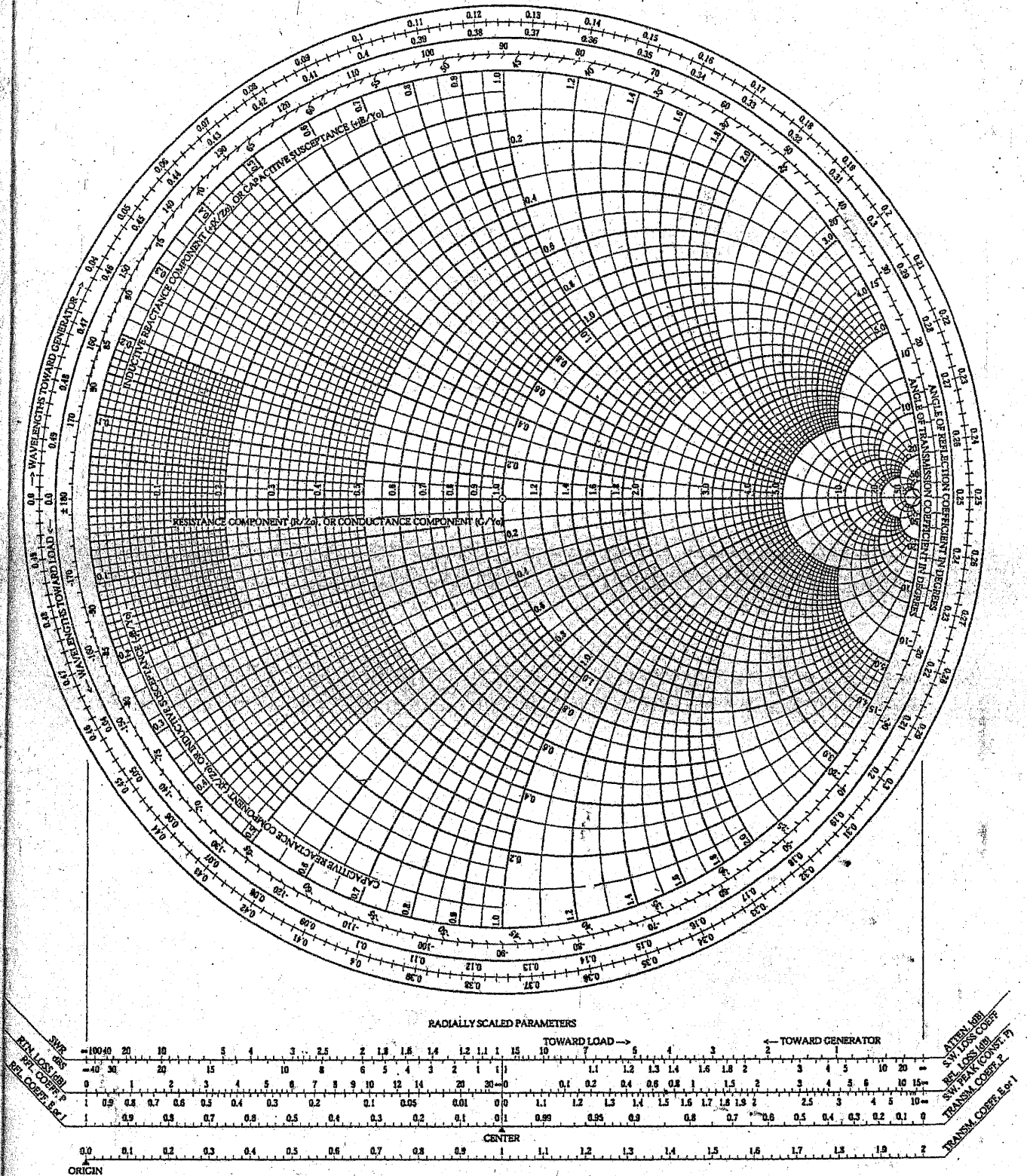
$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{Tmax} = \left(\frac{1 - |\Gamma_S|^2}{|1 - S_{11}\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

The Complete Smith Chart

Black Magic Design



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INSTITUTE OF ENGINEERING
Examination Control Division
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Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- ✓ Necessary figures and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. Differentiate the behaviors of the systems at microwave and conventional low frequency bands. [6]
2. Describe how TE mode is different from TM mode in a circular waveguide. [10]
3. Describe the working principle of a cavity magnetron. [10]
4. Why S-parameter is important in microwave network analysis? Define S-parameters for a two-port network. [4+5]
5. By arbitrarily assuming a suitable load that connects to a 50-ohm transmission line find the lengths and spacing for a two-stub impedance matching system. Assume also a suitable separation between the stubs. [10]
6. Using the following S-parameters of $S_{11}=0.55\angle-150^\circ$, $S_{12}=0.04\angle20^\circ$, $S_{21}=2.82\angle180^\circ$ and $S_{22}=0.45\angle-30^\circ$, calculate and compare maximum power gain for unilateral and bilateral modes. [15]
7. Discuss the difference between an amplifier circuit and an oscillator circuit in terms of stability factor. [5]
8. Write short notes (Any THREE) [3 x 5]
 - a. Microwave magic tee
 - b. Microwave radiation fields
 - c. Microwave strip-lines against micro-strips
 - d. Static calorimeter

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}||S_{21}|}$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where :}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{Tmax} = \left(\frac{1 - |\Gamma_S|^2}{|1 - S_{11}\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

Exam.	Regular		
Level	BE	Full Marks	80
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Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- ✓ Assume suitable data if necessary.

- Based on operational principles, compare microwave systems with conventional low frequency systems. Lists the areas of application of microwave systems. [4+4]
- What makes S-parameters useful in microwave network analysis? Define S-parameters for a two-port network. Justify that the Butterworth and Chebyshev filter responses are common to prototype microwave two-port filter network using insertion loss method. [4+4+4]
- Design a double-stub impedance matching network for a given load of $80 + j180$ Ohm connected to a 100-Ohm transmission line at 3 GHz with a three-eighths wavelength separation between the stubs. Illustrate necessary diagrams to show physical connections. [8+2]
- Define expressions for various field components of a rectangular waveguide in TE mode. Show that a 1 GHz signal cannot propagate in TE₁₀ made in a rectangular waveguide with a wall separation of 5 cm. [7+3]
- Find the maximum gain for a microwave transistor amplifier with $S_{11} = 0.656 \angle 146.7^\circ$, $S_{12} = 0.122 \angle 46.1^\circ$, $S_{21} = 2.3 \angle 44.7^\circ$, $S_{22} = 0.172 \angle -117.1^\circ$. [10]
- What is bunching effect? Briefly describe the construction and operational features of a cavity magnetron. [2+8]
- Describe how standing waves and microwave powers are measured with VSWR meter and bolometry respectively. [4+6]
- Write short notes on: (any two) [2×5]
 - Mixer theory
 - Circulators
 - Microwave radiation hazards and safety practices

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|}$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

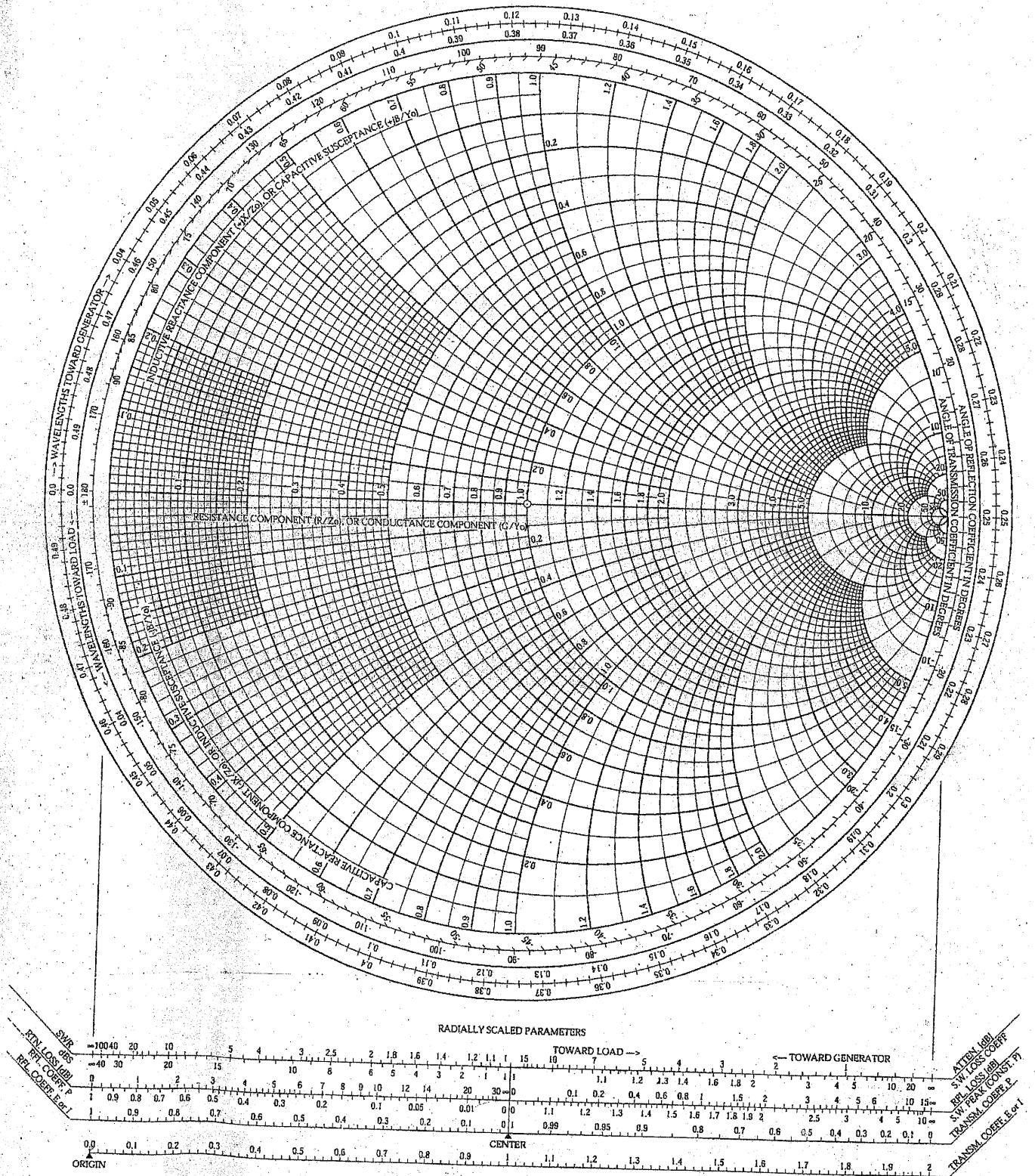
$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{rmax} = \left(\frac{1}{1 - |\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

The Complete Smith Chart

Black Magic Design



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Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and Smith charts are supplied herewith.
- ✓ Assume suitable data if necessary.

1. Differentiate between conventional low frequency and microwave systems based on their working principles. Lists the area of application of microwave systems. [4+4]
2. Justify that S-parameters are used in microwaves instead of h-parameters for network analysis. Define S-parameter for a two-port network. Why the Butterworth and Chebyshev filter responses are common to prototype microwave two-port filter network insertion loss method? [4+4+4]
3. Design a double-stub impedance matching network for a given load of $190 + j110$ Ohms connected to be 100-Ohm transmission line at 10 GHz with a three-eighth wavelength separation between the stubs. Illustrate necessary diagrams to show physical connections. [8+2]
4. Define expressions for various field components of a rectangular waveguide in TM mode. Prove that TM_{01} and TM_{10} modes do not exist in a rectangular waveguide. [7+3]
5. Justify that a transistor having following S-parameters $S_{11} = 0.894 \angle -60.6^\circ$, $S_{12} = 0.020 \angle 62.4^\circ$, $S_{21} = 3.122 \angle 123.6^\circ$ and $S_{22} = 0.781 \angle -27.6^\circ$ is conditionally stable while designing an amplifier. [10]
6. What is transit time effect? Briefly describe the construction and principle of operation of a two-cavity klystron amplifier. [2+8]
7. What is calorimetry in microwave? Differentiate between circulating and flow calorimetries based on principles of operation. [2+8]
8. Write short notes: (any two) [2+8]
 - a) Hybrid tee
 - b) Microwave oscillator theory
 - c) RF radiation hazards and safety standards

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|},$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{T_{\max}} = \left(\frac{1}{1 - |\Gamma_s|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$